

ECON 3200 Discussion Note - Sep 2, 2026

Review: Log Functions

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1 What Is a Logarithm?

The natural logarithm $\ln x$ is the inverse of the exponential function e^x .

That is,

$$y = e^x$$

means exactly the same thing as

$$\ln y = x.$$

2 Three Log Rules

These are the three rules we will need throughout this course.

Product Rule

$$\ln(ab) = \ln a + \ln b$$

Quotient Rule

$$\ln\left(\frac{a}{b}\right) = \ln a - \ln b$$

Power Rule

$$\ln(a^b) = b \ln a$$

The important idea is:

Taking logs turns multiplication into addition and exponents into coefficients.

Example

Suppose

$$y = 5x^2z^{-3}.$$

Take logs:

$$\ln y = \ln(5x^2z^{-3}).$$

Using the product rule,

$$\ln y = \ln 5 + \ln(x^2) + \ln(z^{-3}).$$

Using the power rule,

$$\boxed{\ln y = \ln 5 + 2 \ln x - 3 \ln z.}$$

A multiplicative relationship has become an additive relationship.

3 Why Use a Log Scale?

Compare

$$100, 200, 300, 400$$

with

$$1, 10, 100, 1000.$$

On an ordinary linear axis, the first sequence is equally spaced because

$$200 - 100 = 300 - 200 = 400 - 300 = 100.$$

On a **log axis**, however, what matters is the distance between the logarithms.

For the first sequence,

$$\ln 200 - \ln 100 = \ln \left(\frac{200}{100} \right) = \ln 2,$$

$$\ln 300 - \ln 200 = \ln \left(\frac{300}{200} \right) = \ln 1.5,$$

and

$$\ln 400 - \ln 300 = \ln \left(\frac{400}{300} \right) = \ln \left(\frac{4}{3} \right).$$

Therefore,

$$\ln 2 > \ln 1.5 > \ln \left(\frac{4}{3} \right),$$

so the gaps become smaller.

For

$$1, 10, 100, 1000,$$

we instead have

$$\ln 10 - \ln 1 = \ln 10,$$

$$\ln 100 - \ln 10 = \ln \left(\frac{100}{10} \right) = \ln 10,$$

and

$$\ln 1000 - \ln 100 = \ln \left(\frac{1000}{100} \right) = \ln 10.$$

All three distances are equal.

Interpretation

On a log scale, equal distances represent equal **proportional changes**, rather than equal absolute changes.

For example, 1, 10, 100, and 1000 are equally spaced because each value is ten times the previous value. The difference between their logs is always $\ln 10$.

By contrast, 100, 200, 300, and 400 are equally spaced on a regular scale, but not on a log scale. Their proportional increases become smaller: 100%, 50%, and about 33%.

4 Linear Relationships

A relationship is linear if it can be written as

$$y = a + bx.$$

Its graph is a straight line.

The slope is

$$b = \frac{dy}{dx},$$

and tells us the change in y associated with a one-unit change in x .

For example, if

$$y = 10 + 2x,$$

then

$$\frac{dy}{dx} = 2.$$

So increasing x by one unit increases y by two units.

5 Making Nonlinear Relationships Linear with Logs

This is one of the main reasons economists use logarithms.

Power Function

Start with

$$y = Ax^b.$$

Take logs:

$$\ln y = \ln A + \ln(x^b),$$

so

$$\ln y = \ln A + b \ln x.$$

This is a straight-line relationship between $\ln y$ and $\ln x$.

This is called a **log-log relationship**.

Exponential Function

Start with

$$y = Ae^{bx}.$$

Take logs:

$$\ln y = \ln A + \ln(e^{bx}),$$

so

$$\ln y = \ln A + bx.$$

This is a straight-line relationship between $\ln y$ and x .

This is a **semi-log relationship**.

6 Linear, Semi-log, and Log-log Graphs

Form	Equation	What is logged?
Linear-linear	$y = a + bx$	neither
Semi-log	$\ln y = a + bx$	y only
Log-log	$\ln y = a + b \ln x$	both x and y

What matters economically is what the slope b means.

7 Interpreting Slopes with Calculus

7.1 Linear-linear

Suppose

$$y = a + bx.$$

Differentiate with respect to x :

$$\frac{dy}{dx} = b.$$

So b tells us how many **units** y changes when x changes by one unit.

7.2 Semi-log

Suppose

$$\ln y = a + bx.$$

Differentiate both sides with respect to x . Using

$$\frac{d \ln y}{dy} = \frac{1}{y}$$

and the chain rule,

$$\frac{d \ln y}{dx} = \frac{1}{y} \frac{dy}{dx}.$$

Therefore,

$$\frac{1}{y} \frac{dy}{dx} = b.$$

Equivalently,

$$\boxed{\frac{dy}{y} = b dx}$$

for a small change.

Because dy/y is approximately the proportional change in y , b tells us the proportional change in y associated with a one-unit change in x .

If

$$b = 0.05,$$

then a one-unit increase in x implies approximately a 5% increase in y .

Exact versus Approximate

Since

$$\ln y = a + bx,$$

we can exponentiate:

$$y = e^a e^{bx}.$$

If x increases by one,

$$\frac{y'}{y} = e^b,$$

where $y' = e^a e^{b(x+1)}$. So the exact percentage change is

$$\boxed{100 \times \frac{y' - y}{y} = 100(e^b - 1)\%}.$$

For small b ,

$$e^b - 1 \approx b,$$

so we usually say approximately

$$100b\%.$$

7.3 Log-log

Suppose

$$\ln y = a + b \ln x.$$

Differentiate:

$$\frac{d \ln y}{dx} = b \frac{d \ln x}{dx}.$$

Therefore,

$$\frac{1}{y} \frac{dy}{dx} = b \frac{1}{x}.$$

Multiply both sides by x :

$$\boxed{\frac{dy}{dx} \frac{x}{y} = b.}$$

But

$$\frac{dy}{dx} \frac{x}{y}$$

is exactly the **elasticity** of y with respect to x (e.g, price elasticity: y is the price, and x is the quantity demanded).

Equivalently,

$$\boxed{b = \frac{d \ln y}{d \ln x}.$$

Therefore:

The slope of a log-log relationship is an elasticity.

8 Example for a log-log relationship

Recall

$$y = x^{-1}.$$

Taking logs gives

$$\ln y = -\ln x.$$

Therefore,

$$\frac{d \ln y}{d \ln x} = -1.$$

So the elasticity is -1 .

If x increases by 10%, then for a small change, y decreases by approximately 10%.

The Same Result Using Calculus Directly

Since

$$y = x^{-1},$$

we have

$$\frac{dy}{dx} = -x^{-2}.$$

Elasticity is

$$\frac{dy}{dx} \frac{x}{y} = (-x^{-2}) \frac{x}{x^{-1}} = -1.$$

9 Why Economists Like Log-log Graphs

Suppose we observe a power relationship

$$y = Ax^b.$$

On an ordinary graph, this is generally curved.

After taking logs,

$$\ln y = \ln A + b \ln x.$$

Now it is a straight line, and the slope has a particularly useful interpretation:

$b = \text{elasticity}.$

This is why log-log graphs are extremely common in economics.

Summary

$y = a + bx \quad \Rightarrow \quad b = \frac{dy}{dx}$

units $\rightarrow$ units

$\ln y = a + b \ln x \quad \Rightarrow \quad b = \frac{d \ln y}{d \ln x}$

units $\rightarrow$ percent

$\ln y = a + b \ln x \quad \Rightarrow \quad b = \frac{d \ln y}{d \ln x}$

percent $\rightarrow$ percent = elasticity